BGO and CsI(Tl) crystals read by 64 SiPMs at room temperature: a simulation of three readouts

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Abstract

We simulate the response of two 48×48 mm^2 scintillating crystals, BGO and CsI(Tl), read through optical grease by an 8×8 array of SiPMs at room temperature and irradiated by a ^{22}Na source. The simulation follows each decay from the gamma interactions in the crystal to the charge recorded by three readouts: the Citiroc 1A and the Radioroc2, two ASICs that shape the signal and store its maximum, and the CAEN DT2745, a waveform digitiser whose records are integrated offline. For each of the six combinations of crystal and readout we determine the trigger, the calibration of the SiPM gain from their dark counts, the energy spectrum, the energy resolution at 511 keV and the number of photoelectrons. The crystals produce 3885 (BGO) and 27 600 (CsI(Tl)) photons per 511 keV deposit, of which 84% reach the SiPMs and 41% (BGO) or 32 to 35% (CsI(Tl)) become photoelectrons. The energy resolution is 10.1, 9.1 and 8.5% FWHM for BGO and 7.4, 6.1 and 6.1% for CsI(Tl) with the Citiroc, the digitiser and the Radioroc2 respectively. The intrinsic resolution of the crystals, taken as 5%, and the statistics of the photoelectrons dominate the width; the readout adds a contribution that is largest for the Citiroc, whose short shaping keeps a small and fluctuating fraction of the light, and smallest for the Radioroc2. The dark counts of the SiPMs at room temperature, about 4 MHz per channel, prevent the calibration of the Radioroc2 at its long shaping time and, if the noise of the digitiser does not grow with its gain, favour a digitiser gain of 4 over unit gain. The operating conditions are proposals of the simulation, to be confirmed by the first measurements.

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1 Introduction

The CRYSP project studies scintillating crystals read by arrays of SiPMs for the detection of 511 keV gammas, with the energy and the position of the interaction as the quantities of interest. The measurements of the coming weeks use two crystals, BGO and CsI(Tl), an array of 64 SiPMs and three readouts: the CAEN DT2745 digitiser, which records the waveform of each channel, and two ASICs from Weeroc, the Citiroc 1A and the Radioroc2, which shape the signal of each channel and store its maximum. The three readouts answer the same questions in different ways and with different limitations, and the measurements alone cannot separate the properties of the crystal from those of the readout.

This note presents the simulation of the six combinations of crystal and readout at room temperature, with the operating conditions that the measurements will use. For each of them it establishes how the acquisition is triggered, how the gain of the SiPMs is calibrated, which energy spectrum is recorded, which energy resolution is reached at 511 keV and how many of the emitted photons become photoelectrons and recorded charge. The comparison of the six cases identifies the contributions to the energy resolution and their origin. The position of the interaction will be the subject of a separate note.

Section 2 describes the set-up and Section 3 the simulation. Section 4 gives the operating conditions of the six cases and Section 5 the analysis common to all of them. Sections 6 and 7 present the results for each crystal and readout, Section 8 compares them and Section 9 concludes.

2 The set-up

Both crystals have a face of 48×48 mm^2; the BGO crystal is 22.4 mm and the CsI(Tl) crystal 37.2 mm deep (Figure 1). They are wrapped in PTFE tape, with an air gap between the crystal and the tape, on every face except the one coupled to the SiPMs. The coupling is a layer of optical grease of refractive index 1.42 followed by the silicone resin window of the SiPM tiles, of index 1.57. The SiPMs are Hamamatsu S14161-6050HS-04 tiles, each holding 4×4 channels of 6×6 mm^2 at a pitch of 6.2 mm; four tiles form the 8×8 array, which covers 50×50 mm^2 and therefore overhangs the crystal by 1 mm on each side. A ^{22}Na source sits on the axis of the crystal, 30 mm from the face opposite to the SiPMs. Each decay emits a 1275 keV gamma and, in 90% of the decays, a pair of back-to-back 511 keV gammas from the annihilation of the positron.

3 The simulation

The simulation proceeds in three stages, each writing a file that the next one reads. The first stage generates the ^{22}Na decays, transports the gammas through the crystal (photoelectric absorption and Compton scattering, with pair production absorbed locally) and converts each energy deposit into scintillation photons. The number of photons is the light yield of the crystal, 7600 photons per MeV for BGO and 54 000 for CsI(Tl) at room temperature, times the deposited energy, multiplied by one Gaussian factor per decay that represents the intrinsic resolution of the crystal: 5% FWHM at 511 keV for both crystals, scaling as the inverse square root of the energy. The photons are emitted with the time distribution of the crystal, two components of

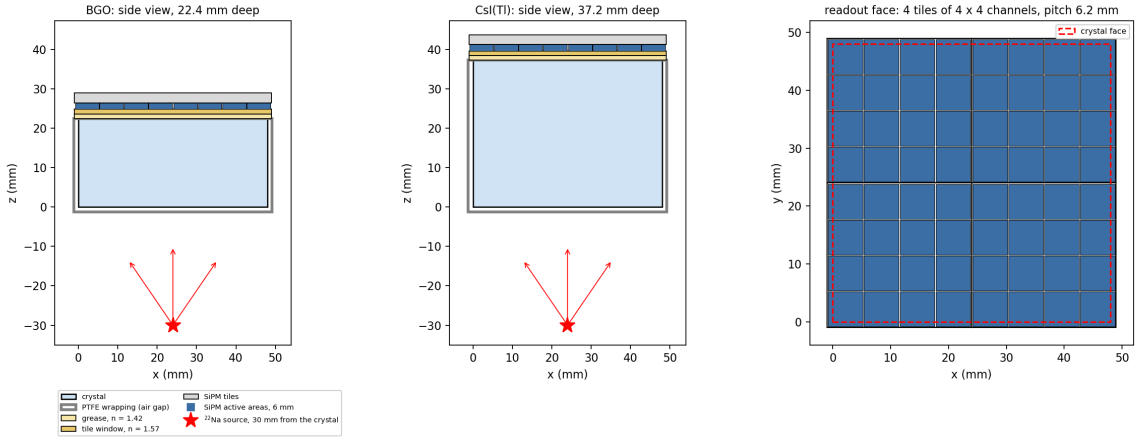


Figure 1: The simulated geometry. Side views of the two crystals with the wrapping, the grease, the window and the SiPM tiles (the coupling layers are drawn thicker than they are), and the source; face view of the 64 active areas, grouped in four tiles, with the outline of the crystal.

300 ns (90%) and 60 ns (10%) for BGO and of 730 ns (64%) and 3340 ns (36%) for CsI(Tl), at the wavelength of the emission maximum, 480 nm for BGO and 550 nm for CsI(Tl). The photons are then transported through the crystal, its surfaces, the wrapping and the coupling layers until they are absorbed or reach the active area of a SiPM; the stage records every photon that arrives there, with its channel, time and wavelength.

The second stage applies the response of the SiPMs. Each recorded photon produces a photoelectron with the photon detection efficiency of the device at its wavelength and at the overvoltage of the run, the excess of the bias voltage over the breakdown voltage. Each photoelectron fires a cell, which may in turn fire neighbouring cells by optical crosstalk and produce delayed afterpulses; independently, thermal dark counts fire cells at random times with the same companions. All these depend on the overvoltage; Table 1 gives their values. We call *fired cells* the sum of photoelectrons, crosstalk and afterpulses produced by the scintillation light, and use the charge of one fired cell, the gain of the SiPM, as the unit of charge: a recorded charge divided by it is expressed in *photoelectron equivalents*. The gain of each channel differs from the nominal one by a Gaussian factor of 5% rms.

The third stage applies the readout to the train of fired cells of each channel. For the Citiroc and the Radioroc2 the train is shaped by a semi-Gaussian filter of order two with the peaking time of the chip, noise is added, a trigger is formed from the discriminators of the chip, and a peak detector stores the maximum of the shaped signal of every channel from the trigger until the *hold*, a fixed delay after the trigger, before the conversion by an ADC. For the digitiser, the charge that the offline analysis integrates over a window, minus the baseline measured before it, is computed directly from the times of the fired cells, and the trigger is formed from the sampled amplitude of each channel with its noise. The three stages and the parameters of each model are described in the design document of the simulation.

4 Operating conditions

The operating conditions are the best compromise found by the simulation for each readout (Table 1). They are proposals for the measurements, and each rests on inputs that the first runs will measure: the gain, crosstalk and dark-count rate of the SiPMs at room temperature, which set the thresholds, and, for the digitiser, its noise at the chosen gain. The simulation assumes the 4 counts per sample of the manual at every gain; if the noise in counts grows in proportion to the gain, the gain of 4 brings no advantage over unit gain. The overvoltage of the SiPMs

is 3 V for both ASICs with BGO; with CsI(Tl), whose light is seven times more abundant, it is lowered to keep the brightest channel within the range of the chip: to 2.7 V for the Citiroc, whose input saturates at 400 pC, and to 2.0 V for the Radioroc2, whose low-gain path saturates at 2000 photoelectrons of unit gain 10^6. The digitiser operates at 5 V, the setting used so far on the bench, at a gain of 4, which places one photoelectron at 64 ADC counts against the assumed noise of 4 counts per sample. The Citiroc shapes the signal with its longest peaking time, 87.5 ns; the Radioroc2 with its longest, 1.8 µs, on both its high-gain and low-gain paths, and the energy is read from the low-gain path. The digitiser integrates over 1 µs for BGO and 3 µs for CsI(Tl), starting 50 ns before the trigger, with a baseline measured over the 900 ns before the integration window; these values minimise the energy resolution in a scan of the window.

Table 1: Operating conditions of the six cases. The detection efficiency is the photon detection efficiency at the peak of its spectral curve; the dark-count rate is per channel; the window is the hold of the ASICs and the integration window of the digitiser. Thresholds in photoelectron equivalents.

	BGO			CsI(Tl)		
	Citiroc	DT2745	Radioroc2	Citiroc	DT2745	Radioroc2
overvoltage (V)	3.0	5.0	3.0	2.7	5.0	2.0
SiPM gain (10^6)	2.8	4.7	2.8	2.5	4.7	1.9
detection efficiency	0.52	0.54	0.52	0.50	0.54	0.41
crosstalk probability	0.088	0.195	0.088	0.070	0.195	0.041
dark-count rate (MHz)	4.3	4.4	4.3	4.1	4.4	3.3
shaping (ns)	87.5	—	1800	87.5	—	1800
window (ns)	300	1000	3560	500	3000	3560
trigger	single, 8	two, 7.5	charge, 40	single, 7	two, 7.5	charge, 40

The trigger differs between the readouts. The Citiroc triggers when one channel crosses the threshold of its fast discriminator, which follows a shaper of 15 ns. The digitiser triggers when two channels cross the threshold within 100 ns, a coincidence that we call a majority of two. The Radioroc2 triggers on its charge discriminator, which reads the low-gain shaped signal of each channel. In each case the threshold is the lowest one at which every 511 keV photopeak event triggers (an efficiency of at least 0.995) while the dark counts, alone, trigger less than once per millisecond. At room temperature this second condition is the restrictive one: a dark count with three or four crosstalk companions occurs in some channel in most microseconds, and thresholds of a few photoelectrons trigger on it.

5 Analysis

The analysis treats the simulated files as it will treat the measured ones, and uses the record of the simulation only to select the events and to follow the light from the crystal to the readout.

Calibration. The gain of each channel is measured on the dark counts, recorded without source. For the ASICs, each channel triggers on its own discriminator at half a photoelectron and its high-gain signal is held at the peaking time after the trigger, which records the amplitude of the dark pulse that triggered; only the channels whose discriminator fired within 2 ns of the trigger enter the spectrum, since the others are held at a random phase of their signal. For the digitiser, every record contains tens of dark pulses per channel; the waveform is filtered with the sampled pulse of one fired cell (a matched filter) after subtracting the mean of the four preceding samples, and the local maxima of the filtered waveform are the pulse amplitudes. In both cases the position of the one-photoelectron peak of each channel, from a Gaussian fit, is its gain.

Pedestal. The peak detectors of the ASICs store, in a channel that receives no light, the largest dark pulse or noise excursion within the hold, never zero. This pedestal is measured on

events with no energy deposit read with a random trigger, and subtracted channel by channel. It amounts to 1.5 photoelectron equivalents per channel for the Citiroc and 9 to 12 for the Radioroc2, whose hold is ten times longer; the digitiser, whose baseline is subtracted, has none.

Energy. The energy of an event is the sum of the 64 channels. A straight line through the positions of the 511 and 1275 keV peaks converts it into keV; this calibration absorbs the residual offset left by the pedestal, which is not exactly additive, since in a channel that receives light a dark pulse changes the stored maximum only when it falls on top of the signal. The 511 keV peak is fitted with a Gaussian on a linear background, and the energy resolution is its FWHM divided by its position.

Light. The events in which one 511 keV gamma deposits its full energy in the crystal, without any deposit of the 1275 keV gamma, are selected from the record of the simulation; they number between 7000 and 52 000 per case. On these events we count the photons emitted, the photons that reach the active areas of the SiPMs, the photoelectrons and the fired cells, the last two from a replay of the SiPM response without dark counts, and we take the recorded charge at 511 keV in photoelectron equivalents from the energy calibration. The ratios between consecutive quantities are the collection efficiency of the crystal and its coupling, the detection efficiency at the emission wavelength, and the fraction of the photoelectrons that the readout records.

Resolution. On the same events, the relative width of each quantity (half the interval between its 16th and 84th percentiles divided by its median, multiplied by 2.355 to compare with a FWHM) separates the contributions to the energy resolution. The width of the number of photoelectrons contains the intrinsic resolution of the crystal and the statistics of the light collection and detection; the difference in quadrature with the intrinsic 5% is the statistical contribution. The difference in quadrature between the widths of the fired cells and of the photoelectrons is the contribution of crosstalk and afterpulses; that between the recorded energy and the fired cells is the contribution of the readout, which includes the dark counts, the electronic noise, the fluctuation of the fraction of the light that the shaping keeps and the pedestal.

6 BGO

A 511 keV deposit in BGO emits 3885 photons, of which 3254 (84%) reach the SiPMs and 1585 become photoelectrons at 3 V, 1634 at 5 V. The width of the number of photoelectrons is 7.7% FWHM, the intrinsic 5% and a statistical term of 5.9%, the value expected from 1585 photoelectrons.

6.1 Citiroc 1A

The single-channel threshold of 8 photoelectron equivalents triggers on every 511 keV event and on 98.5% of the deposits between 200 and 300 keV, while the dark counts trigger at 0.3 kHz (Figure 2). The single-photoelectron spectrum shows clean peaks; the fitted gain is 0.885 of the true one, because the hold at the peaking time after the discriminator falls slightly after the maximum, a factor common to all channels that the energy calibration absorbs. The fitted gains follow the true ones channel by channel to 1.4%, against the 5% spread of the gains; in 6 channels of 64 the fit takes the two-photoelectron peak and must be repeated. The shaping of 87.5 ns keeps a small part of a 300 ns light pulse: the recorded charge at 511 keV is 726 photoelectron equivalents, 0.46 of the photoelectrons. The energy resolution is 10.1% FWHM. The readout contributes 5.9%, as much as the statistics: the fraction of the light that a short shaping keeps depends on the times of arrival of the photoelectrons and fluctuates from event to event. The ratio of the 1275 and 511 keV peak positions is 2.48 before calibration, close to the 2.50 of a linear response.

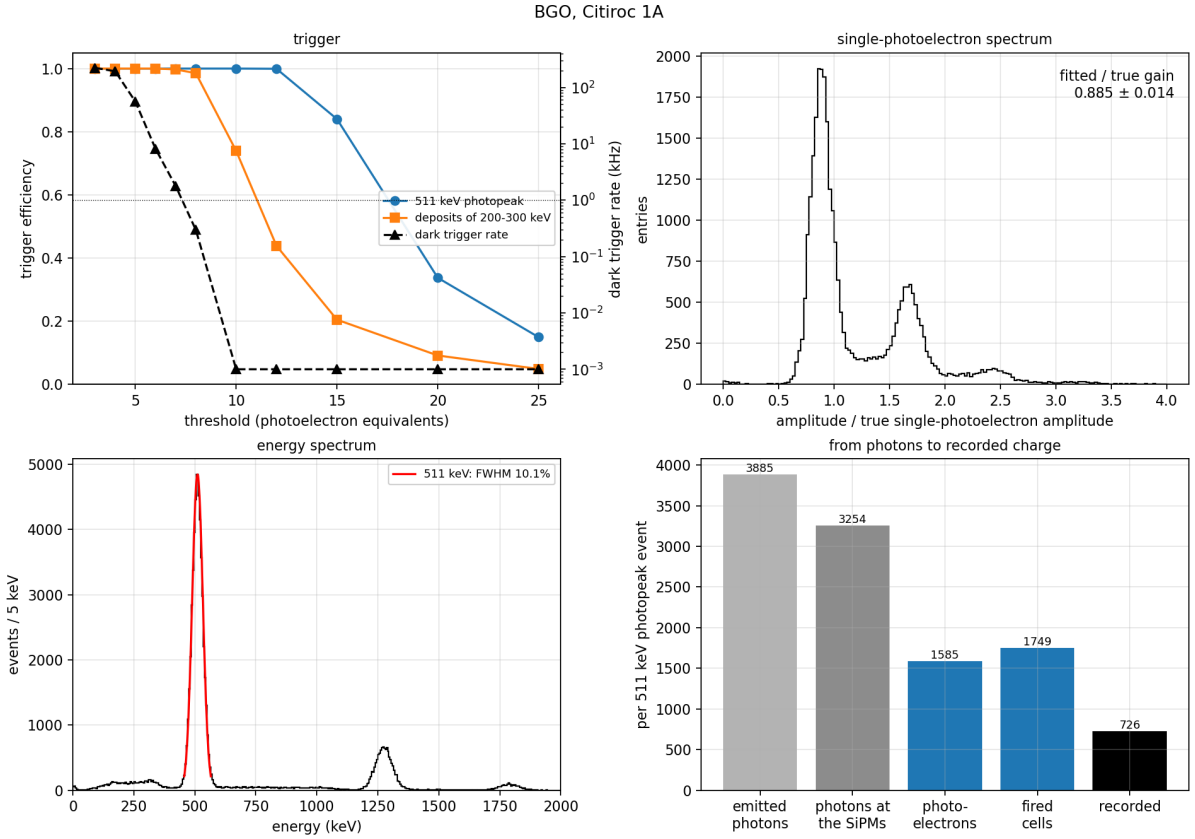


Figure 2: BGO read by the Citiroc 1A. Top left: trigger efficiency for the 511 keV photopeak and for deposits of 200 to 300 keV, and dark trigger rate, against the threshold; rates below the sensitivity of the sample are drawn at 10^{-3} kHz. Top right: single-photoelectron spectrum of the dark counts, all channels, each normalised to its true gain. Bottom left: calibrated energy spectrum with the fit of the 511 keV peak. Bottom right: photons emitted, photons at the SiPMs, photoelectrons, fired cells and recorded charge per 511 keV photopeak event.

6.2 DT2745 digitiser

The threshold of 480 ADC counts, 7.5 photoelectrons, with a majority of two triggers on every 511 keV event and on 98.6% of the deposits between 200 and 300 keV; no event without deposit triggers in the sample, which bounds the dark trigger rate below 0.3 kHz (Figure 3). At gain 4 the matched filter separates the single-photoelectron peaks cleanly; the fitted gain is 0.981 of the true one with a spread of 0.1% between channels, and the one-photoelectron peak has a width of 6% of its position. The integration window of 1 µs records the whole light pulse, so that the recorded charge, 1966 photoelectron equivalents, exceeds the 1634 photoelectrons by the crosstalk (19.5% at 5 V) and the afterpulses. The energy resolution is 9.1% FWHM. Crosstalk and afterpulses contribute 2.9%, and the readout 4.3%: the integration window collects 280 dark counts on average over the 64 channels, with their crosstalk, and their fluctuation adds to the width. The ratio of the peak positions is 2.49.

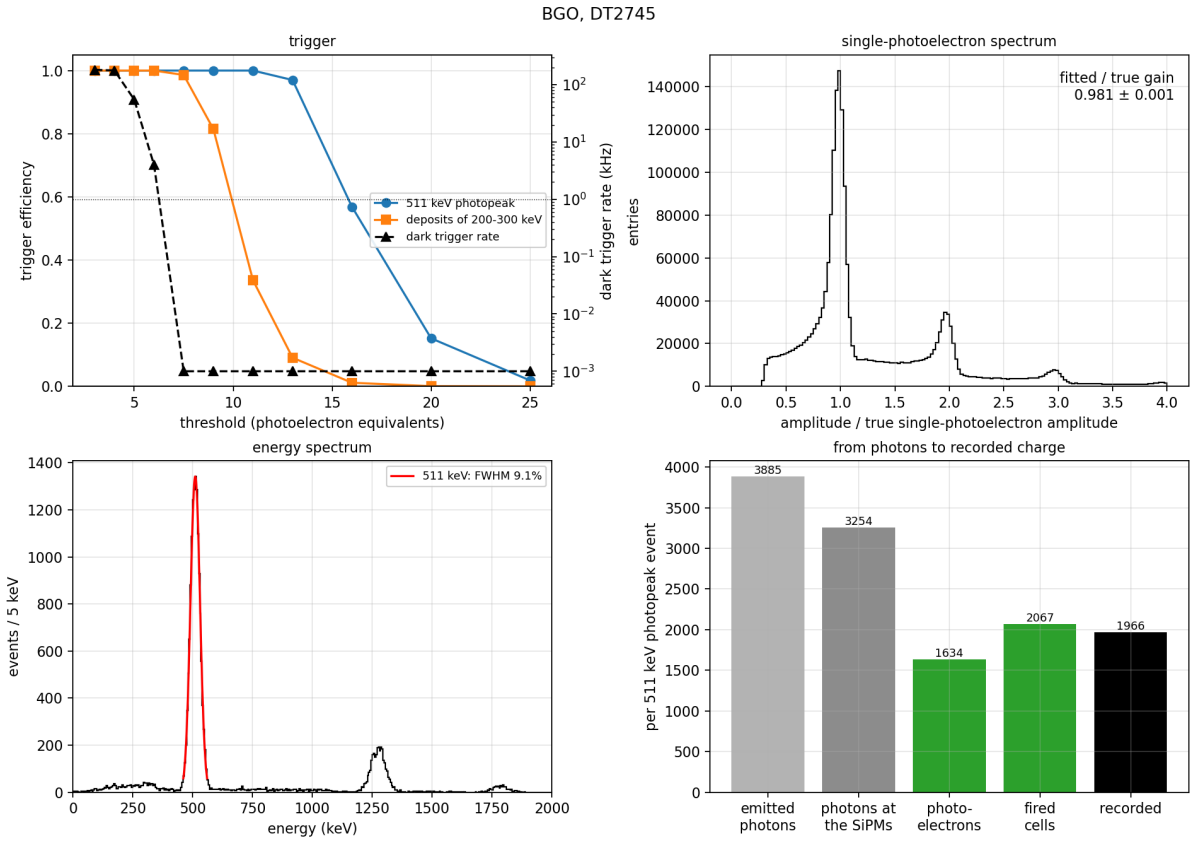


Figure 3: BGO read by the DT2745 digitiser at gain 4 and 5 V of overvoltage. Panels as in Figure 2; the single-photoelectron spectrum is that of the matched-filter amplitudes.

6.3 Radioroc2

The charge trigger at 40 photoelectron equivalents triggers on every 511 keV event but only on 72% of the deposits between 200 and 300 keV, since it requires one channel above threshold, and the brightest channel of such a deposit is often below it; dark counts do not trigger (Figure 4). At the shaping of 1.8 µs no single-photoelectron peak exists: about eight dark counts fall within the shaping time of each channel and their signals pile up. The calibration is therefore taken at a peaking time of 300 ns, where the peaks are clean and the fitted gain follows the true one to 1.4%; the gain at 1.8 µs follows from the ratio of the shaper gains at the two peaking times, which the simulation knows exactly and the measurement must determine, for instance with the

charge-injection input of the chip. The long shaping keeps almost all the light: the recorded charge, 1670 photoelectron equivalents, is 1.05 times the photoelectrons. The pedestal is 785 photoelectron equivalents over the 64 channels; its subtraction leaves an offset of 64 keV in the energy scale, and a ratio of the peak positions of 2.71, which the two-point calibration absorbs. The energy resolution is 8.5% FWHM, with a readout contribution of 2.9%.

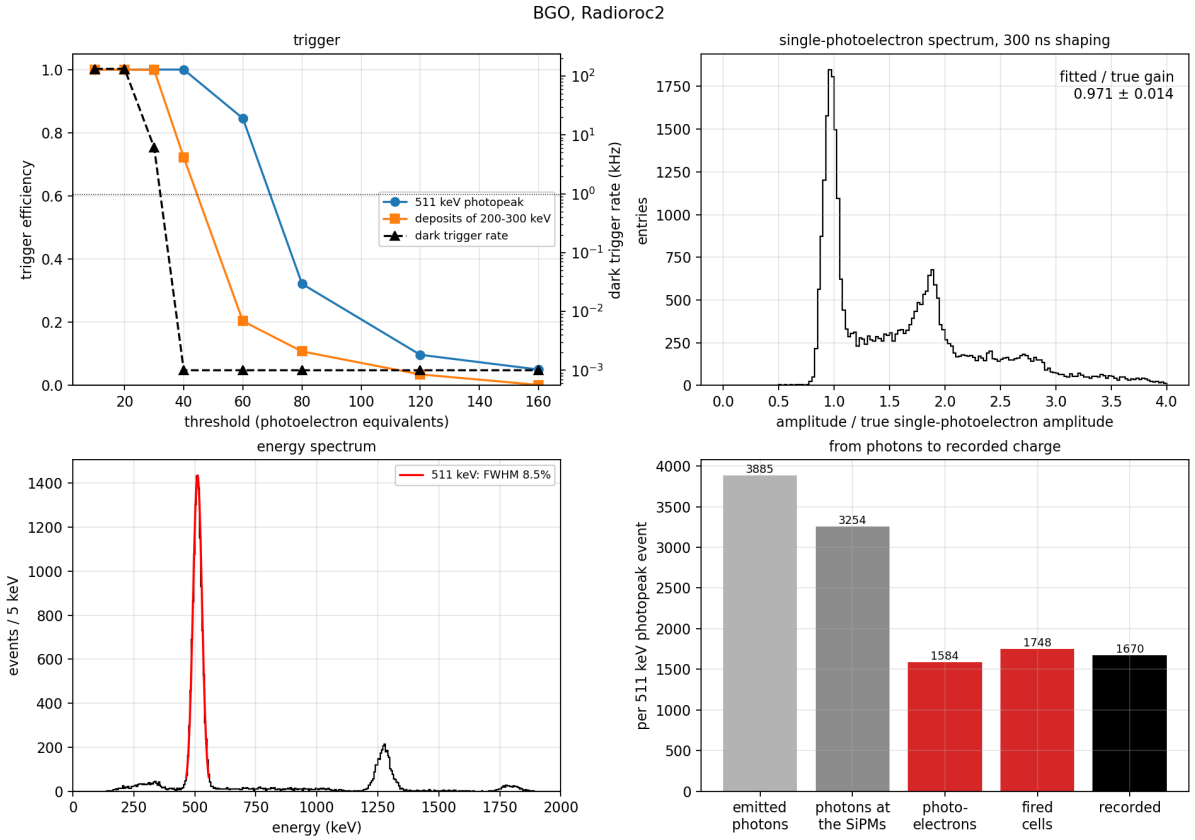


Figure 4: BGO read by the Radioroc2 at 1.8 µs of shaping. Panels as in Figure 2; the single-photoelectron spectrum is taken at 300 ns of shaping.

7 CsI(Tl)

A 511 keV deposit in CsI(Tl) emits 27 600 photons, of which 23 210 (84%) reach the SiPMs. The detection efficiency at the 550 nm emission is lower than at 480 nm, and the number of photoelectrons depends on the overvoltage: 7215 at 2.0 V, 8820 at 2.7 V and 9523 at 5 V. The statistical contribution to the width is therefore 2.4 to 3.0%, half of the intrinsic 5%.

7.1 Citiroc 1A

The single-channel threshold of 7 photoelectron equivalents triggers on every event of the two classes, at a dark trigger rate of 0.6 kHz (Figure 5); the calibration is that of BGO, with a fitted gain of 0.884 of the true one and a spread of 1.0% between channels. The shaping of 87.5 ns keeps a sixth of the photoelectrons of a light pulse that lasts microseconds: the recorded charge is 1461 photoelectron equivalents, 0.17 of the 8820 photoelectrons. The energy resolution is 7.4% FWHM, of which the readout contributes 5.0%, the largest term. The brightest channel exceeds the input range of the chip in 3.4% of the photopeak events; this saturation bends the response, with a ratio of the peak positions of 2.44.

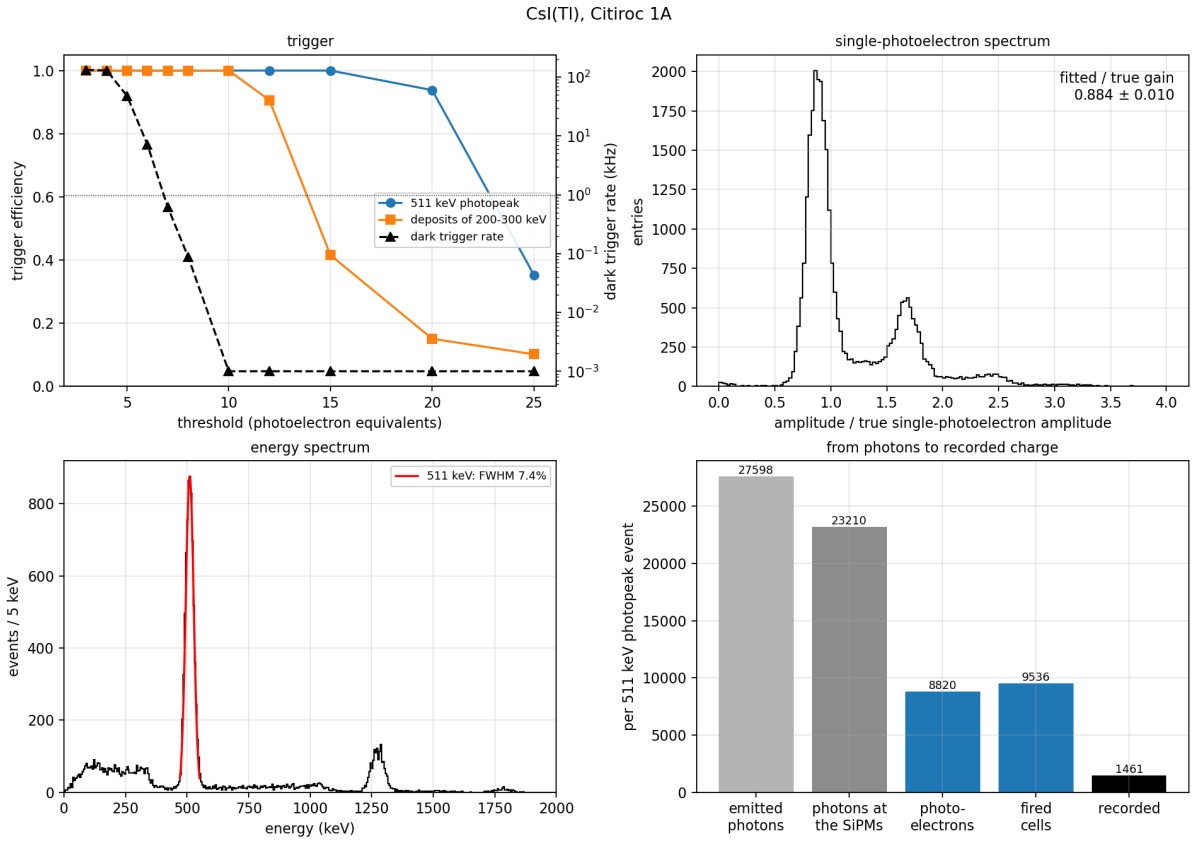


Figure 5: CsI(Tl) read by the Citiroc 1A. Panels as in Figure 2.

7.2 DT2745 digitiser

The trigger and the calibration are those of BGO; every event of the two classes triggers and no dark trigger occurs in the sample (Figure 6). The integration window of 3 µs holds most of both light components: the recorded charge is 10 087 photoelectron equivalents, 1.06 times the 9523 photoelectrons, the crosstalk compensating the light that falls after the window. The energy resolution is 6.1% FWHM. The window collects 850 dark counts on average, and the readout contributes 2.7%; no channel saturates.

7.3 Radioroc2

The charge trigger at 40 photoelectron equivalents triggers on every event of the two classes (Figure 7). The calibration at 300 ns gives a fitted gain of 0.967 of the true one, with a spread of 1.5%. At 2.0 V the brightest channel exceeds the low-gain range in 0.7% of the photopeak events, against at least 2.7% at 2.7 V, at the price of 18% fewer photoelectrons than at 2.7 V. The shaping of 1.8 µs keeps 0.74 of the photoelectrons, 5331 photoelectron equivalents. The energy resolution is 6.1% FWHM, with a readout contribution of 1.8%, the smallest of the six cases; the ratio of the peak positions after the pedestal subtraction is 2.52.

8 Comparison

Figure 8 gathers the energy resolutions and their decomposition. For BGO the three readouts span 8.5 to 10.1%, for CsI(Tl) 6.1 to 7.4%, and in every case the intrinsic resolution of the crystal and the statistics of the photoelectrons account for most of the width. For BGO the statistical term, 5.9%, is as large as the intrinsic one, because the crystal produces few photons;

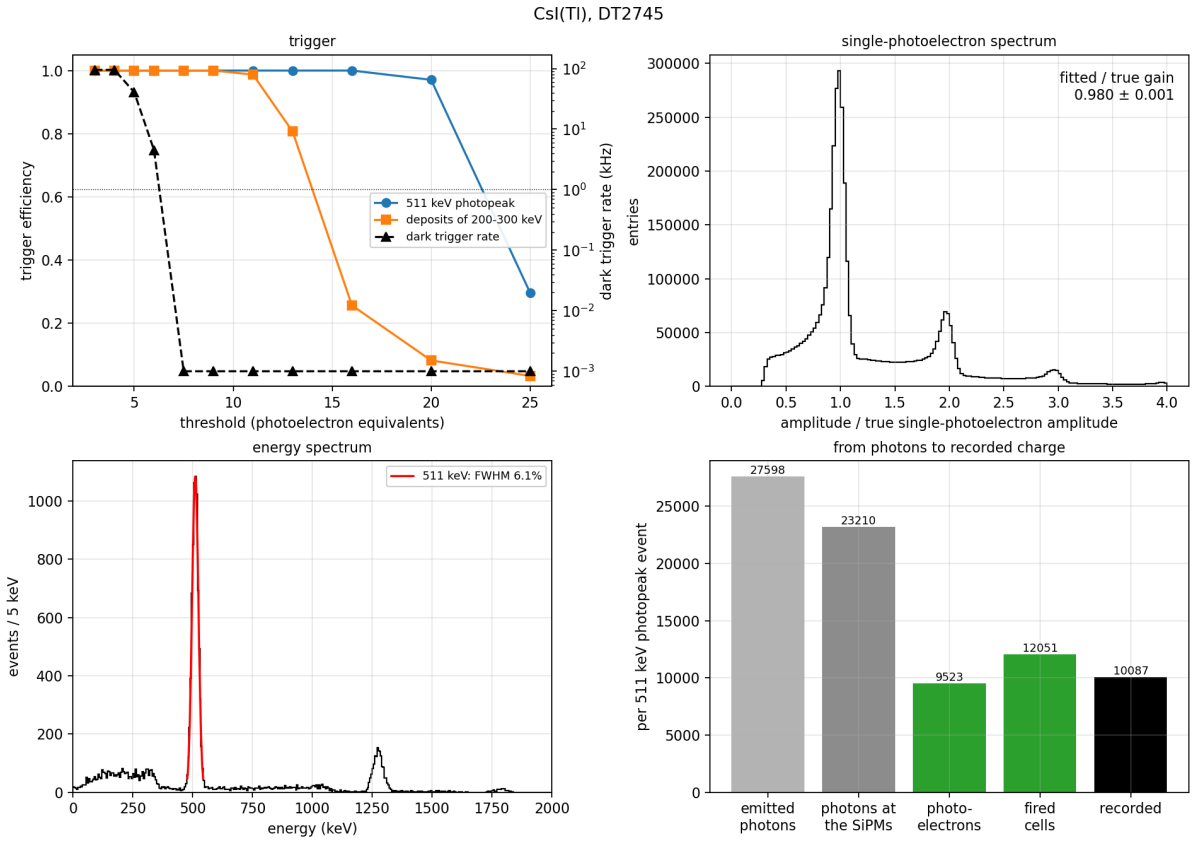


Figure 6: CsI(Tl) read by the DT2745 digitiser. Panels as in Figure 3.

for CsI(Tl) it is half of it. The readout term orders the three readouts in the same way for both crystals: it is largest for the Citiroc, whose shaping of 87.5 ns records a fluctuating fraction of a long light pulse; intermediate for the digitiser, whose integration window collects dark counts in proportion to its length; and smallest for the Radioroc2, whose 1.8 µs shaping records nearly all the light. The digitiser pays in addition for the 19.5% crosstalk of the 5 V overvoltage at which it operates. With BGO, the digitiser at 3 V of overvoltage, where the crosstalk is 9%, reaches 8.8% in the same simulation; with CsI(Tl) it makes no difference.

Figure 9 shows the chain from photons to recorded charge. The collection efficiency, 0.84, is a property of the crystal and its coupling and is common to the three readouts. The detection efficiency at the emission wavelength is 0.49 to 0.50 for BGO and 0.31 to 0.41 for CsI(Tl), depending on the overvoltage. The fraction of the photoelectrons recorded spans an order of magnitude: 0.17 for CsI(Tl) and 0.46 for BGO with the Citiroc, close to one with the Radioroc2 and above one with the digitiser, whose integral includes the crosstalk. The number of photoelectrons sets the statistical limit; the recorded fraction sets how close a readout comes to it.

Figure 10 shows the distributions of the brightest and of the least illuminated channel in the photopeak events, the two ends of the dynamic range that a position measurement needs. In BGO the brightest channel holds a median of 29 photoelectron equivalents with the Citiroc, 58 with the Radioroc2 and 74 with the digitiser, and the least illuminated one 3 to 9; in CsI(Tl) the brightest channel reaches 1800 photoelectron equivalents in one event of a hundred with the digitiser, which its range holds, and 940 with the Radioroc2, at the limit of its low gain.

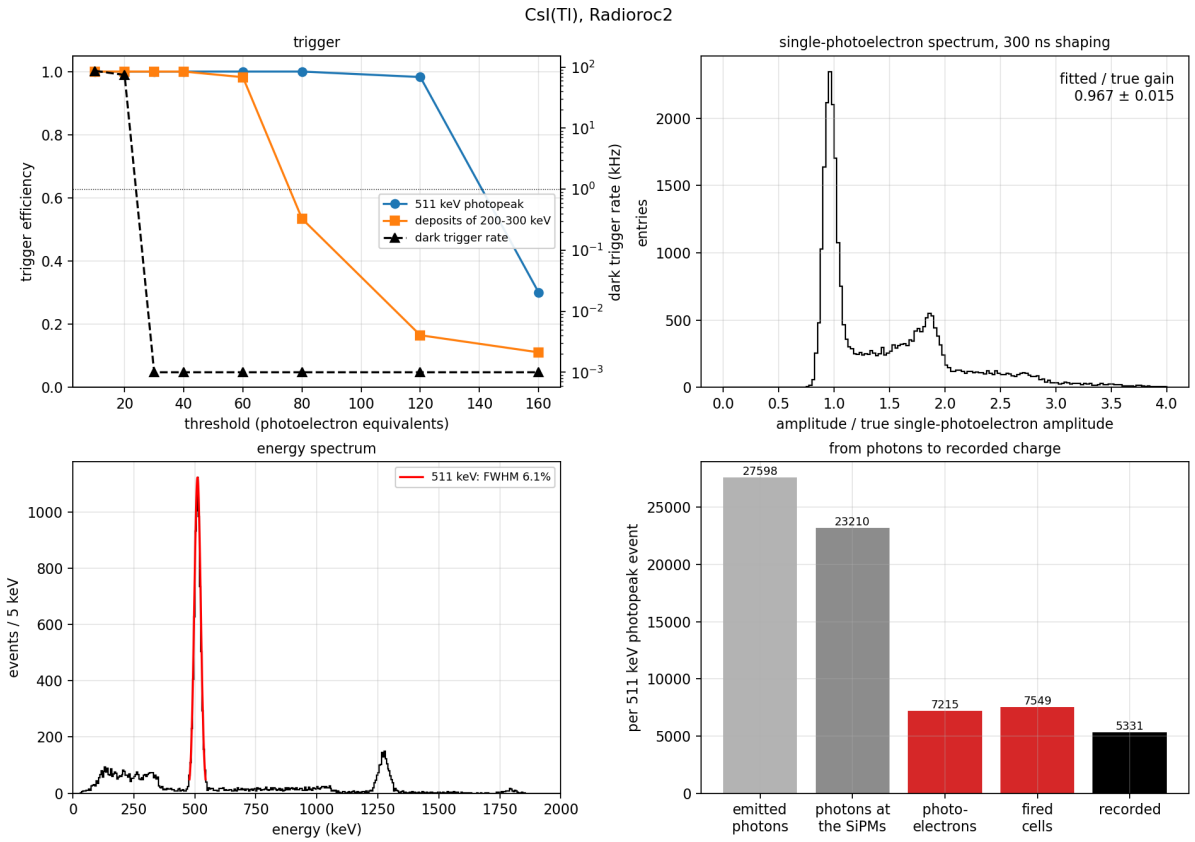


Figure 7: CsI(Tl) read by the Radioroc2. Panels as in Figure 4.

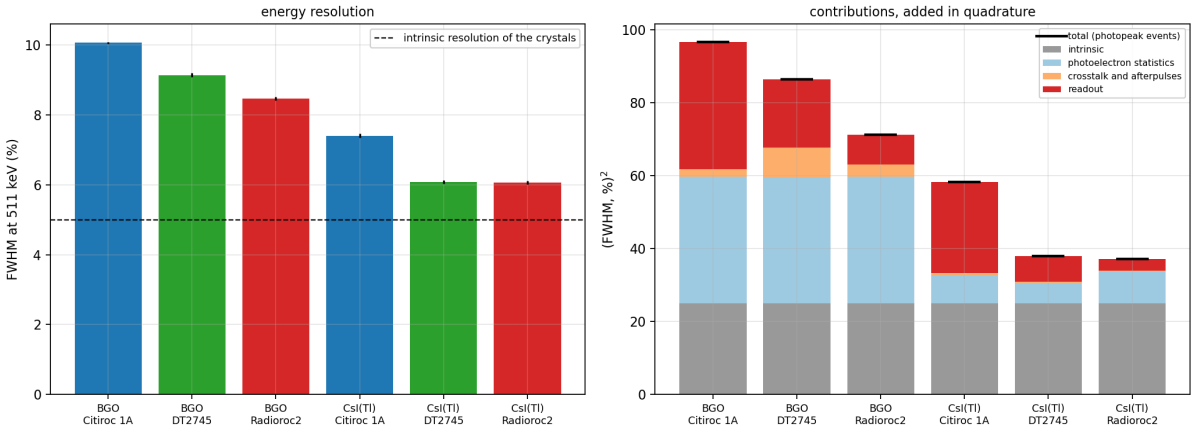


Figure 8: Left: energy resolution at 511 keV of the six cases, with the intrinsic resolution of the crystals. Right: the squares of the contributions to the width of the photopeak events, stacked; the bar marks the total.

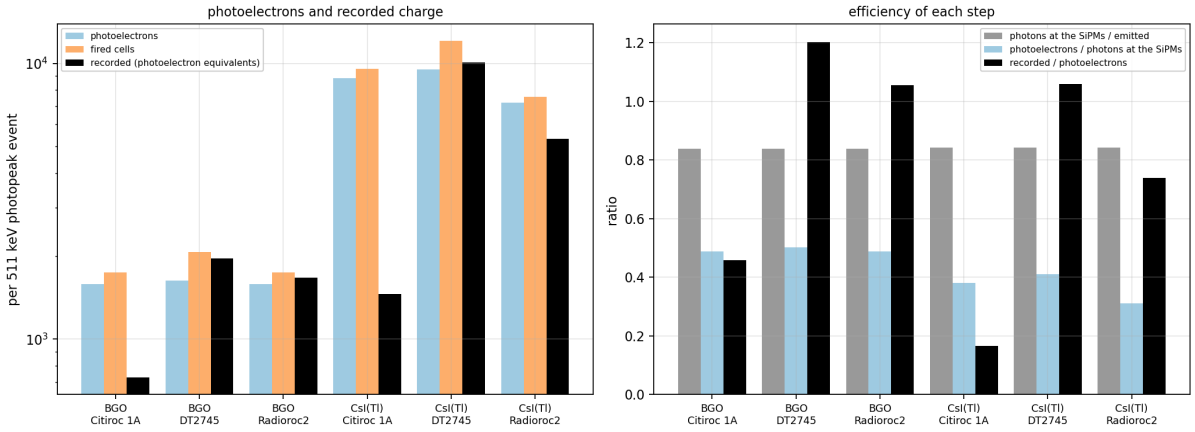


Figure 9: Left: photoelectrons, fired cells and recorded charge per 511 keV photopeak event. Right: collection efficiency, detection efficiency and fraction of the photoelectrons recorded.

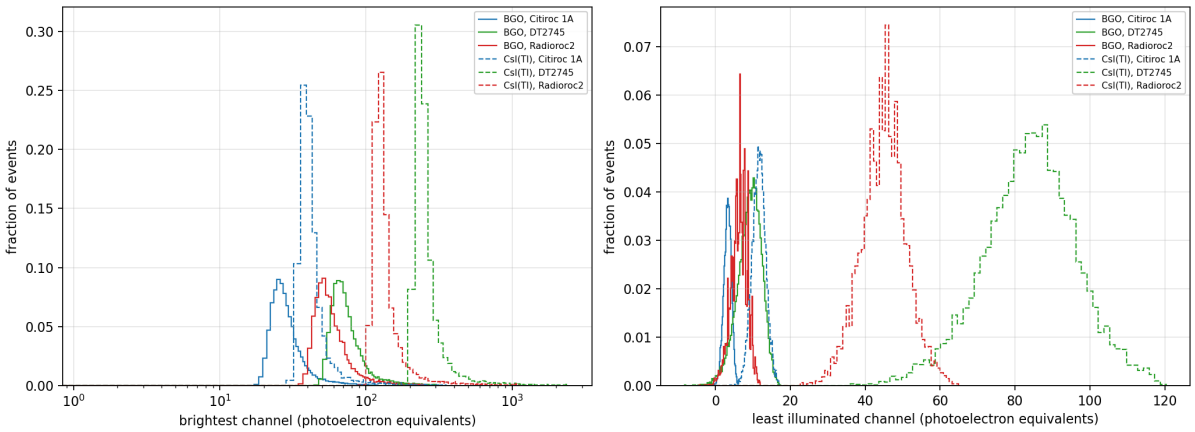


Figure 10: Brightest (left) and least illuminated (right) channel of the 511 keV photopeak events, pedestal subtracted, in photoelectron equivalents. Solid lines BGO, dashed lines CsI(Tl).

9 Conclusions

At room temperature the three readouts measure the energy of a 511 keV deposit with resolutions within 1.6 points of each other for BGO and 1.3 points for CsI(Tl). The resolution is limited by the crystal, through its intrinsic resolution and, for BGO, its low light yield; the readout adds a term that ranges from 5 to 6% FWHM for the Citiroc down to 2 to 3% for the Radioroc2. The dark counts of the SiPMs, about 4 MHz per channel, shape the operation of every readout: they set the trigger thresholds at 7 to 8 photoelectrons, add to the width through the integration window of the digitiser and the pedestal of the peak detectors, and determine how the gain can be calibrated. In the simulation the digitiser separates its single-photoelectron peaks at gain 4, provided its noise in counts does not grow with the gain, and only with a matched filter; the Radioroc2 must be calibrated at a short shaping time and transfer the gain to its long one; the Citiroc calibrates directly. The first measurements decide whether these proposals stand: the noise of the digitiser at gains 1 and 4, the single-photoelectron spectra, the dark-count rate and crosstalk of the SiPMs and the trigger rate against the threshold, and then the number of photoelectrons, which is the least dependent on the readout.